

TEXT version of Transcript

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* John F. Fairbanks

Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer

* Laura J. Guerrant-Oiye

Aspen Aerogels, Inc. - Vice President of Investor Relations & Corporate Communications

* Ricardo C. Rodriguez

Conference Call Participants

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* Eric Andrew Stine

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Presentation

Operator [1]

Good afternoon. Thank you for attending the Aspen Aerogels Inc. Fourth Quarter 2021 Earnings Call.
[Operator Instructions]

I would now like to turn the conference over to your host, Laura Guerrant, with Aspen Aerogels. Thank you. You may proceed, Ms. Guerrant.

Laura J. Guerrant-Oiye, Aspen Aerogels, Inc. - Vice President of Investor Relations & Corporate Communications [2]

Thank you, Lauren.

Good evening, and thank you for joining us for the Aspen Aerogels Fiscal Year 2021 Financial Results Conference Call. I'm Laura Guerrant, Aspen's Vice President of Investor Relations and Corporate Communications. With us today are Don Young, President and CEO; John Fairbanks, our Chief Financial Officer, and Ricardo Rodriguez.

Ricardo is Aspen's Chief Strategy Officer and he will be assuming the role of CFO upon John's retirement April 1. There are a few housekeeping items that I'd like to address before turning the call over to John. The press release announcing Aspen's financial results and business developments as well as a reconciliation of management's use of non-GAAP financial measures compared to the most applicable GAAP measures is available on the Investors section of Aspen's website www.aerogel.com.

Included in the press release is a summary statement of operations, a summary of balance sheet and a summary of key financial and operating statistics for the fourth quarter and full-year ended December 31, 2021. In addition, I'd like to highlight that we have uploaded to our website a slide deck that will accompany our conversation today. You can find the deck at the Investors section of our website. An archive of today's webcast will be on our website for approximately one year.

Please note that our discussion today will include forward-looking statements including any statements regarding outlook, expectations, beliefs, projections, estimates, targets, prospects, business plans and any other statement that is not a historical fact. These forward-looking statements are subject to risks and uncertainties. Aspen Aerogels' actual results may differ materially from those expressed in these forward-looking statements.

A list of factors that could affect the company's actual results can be found in Aspen's press release issued today, Page 1 of the presentation and as discussed in more detail on the reports Aspen files with the SEC, particularly in the company's most recent Annual Report on Form 10-K.

The company's press release issued today and filings with the SEC can also be found on the Investors section of Aspen's website. Forward-looking statements made today represent the company's views as of today, February 17, 2022. Aspen Aerogels disclaims any obligation to update these forward-looking statements to reflect future events or circumstances.

During this call, we will refer to non-GAAP financial measures including adjusted EBITDA. These financial measures are not prepared in accordance with US Generally Accepted Accounting Principles or GAAP. These non-GAAP financial measures are not intended to be considered in isolation or as a substitute for results prepared in accordance with GAAP. The definitions and reconciliations of these non-GAAP financial measures to the most directly comparable GAAP financial measures and a discussion of why we present these non-GAAP financial measures are included in today's press release.

I'll now turn the call over to John. John?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [3]

Thanks, Laura.

I'll provide a summary of our 2021 financial results and discuss our 2022 outlook before turning the call over to Don and Ricardo.

At a high level, our 2021 results and our 2022 outlook share the same drivers. First, we're generating revenue growth in each of our principal markets. We're benefiting from the rapid adoption of our PyroThin thermal barriers in the electric vehicle market. We're experiencing solid growth in the energy infrastructure market associated with the initial stages of a post-COVID recovery and we're gaining share in the European sustainable building materials market with our spacelin -- Spaceloft line products.

Second, we're investing in people, resources and capital assets to support our rapidly growing e-mobility opportunities. This investment includes increased spending to enhance our technical, operational, and commercial teams supporting our thermal barrier business, the research and development team supporting our carbon aerogel battery material opportunity, our legal resources to expand and defend our IP portfolio and our finance, information technology and general management personnel to manage the anticipated strong growth in our business.

This investment also includes planned capital expenditures to design and construct our second silica aerogel manufacturing plant, to build out our thermal barrier fabrication operations and to expand our carbon aerogel battery materials production, fabrication and testing facilities. We're confident that these investments are commensurate with the scale of our EV opportunity.

Focusing in on 2021, total revenue increased by \$21.3 million versus 2020 to \$121.6 million. This \$21.3 million increase in revenue was composed of a \$12.6 million increase in energy infrastructure revenue, \$6.7 million in initial revenue in the EV thermal barrier market and \$1.9 million of growth in the sustainable building materials market. Our revenue could have been higher.

We estimate that our production output was depressed by between 6 and \$8 million during the fourth quarter due to COVID-related staffing issues and some raw material shortages. Without these issues, our 2021 revenue would have been in the upper end of our outlook range, which as a reminder, we raised three times during the year. Importantly, we are experiencing an improvement in both staffing and raw material availability thus far during the first quarter of 2022.

Turning to costs and expenses. During 2021, material costs increased by 33% or \$14.5 million, manufacturing expenses by 28% or \$11.6 million and operating expenses by 40% or \$14.3 million versus 2020. The growth in cost and expenses were driven by both the increase in revenue by our investment in people and resources to prepare for growth in 2022 and beyond. Accordingly, net loss increased to \$37.1 million or \$1.22 per share in 2021 versus a net loss of \$21.8 million, or \$0.83 per share in 2020. And adjusted EBITDA was negative \$26 million in 2021 compared to negative \$6.4 million in 2020.

As a reminder, we define adjusted EBITDA as net income or loss before interest, taxes, depreciation, amortization, stock-based compensation expense and any other items that we do not believe are indicative of our core operating performance. In 2021, these other items included a \$3.7 million gain on extinguishment of debt.

Next, I'll turn to our balance sheet and cash flow for 2021. Cash used in operations of \$18.6 million reflected our adjusted EBITDA of negative \$26 million, offset in part by a \$7.4 million decrease in working capital investment. Capital expenditures during the year of \$13.8 million included engineering and designs for our second manufacturing facility and investments to expand our carbon aerogel capacity.

Cash provided by financing activities of \$92.5 million included \$73.5 million of proceeds from our June 2021 private placement with Koch strategic partners and \$19.4 million from sales of equity through our ATM facility. We ended the year with \$76.6 million of cash, no borrowings under our revolving credit facility and shareholders' equity of \$128.4 million. We also had \$12.6 million available under our revolving credit facility at year-end.

I'll now turn to our full-year 2022 outlook. We expect continued revenue growth in 2022, associated with a tripling of our PyroThin thermal barrier revenue to \$20 million for the year, the continuation of the post-

COVID recovery in the global energy infrastructure market and continued demand growth in the European sustainable building materials market. As a result, we're setting our 2022 revenue outlook to between 145 and \$155 million for the year equivalent to growth of between 19% and 27% compared to 2021.

As discussed earlier, we plan to continue to increase investment in personnel and resources to capitalize on our rapidly growing e-mobility opportunities. Accordingly, our initial 2022 full-year outlook is as follows. We expect total revenue of between 145 and \$155 million, net loss of between 66.7 and \$70.7 million, adjusted EBITDA of between negative \$42 million and negative \$46 million, EPS of between a loss \$1.95 and a loss of \$2.07 per share.

This EPS outlook assumes a weighted average of 34.2 million shares outstanding for the year. In addition, this 2022 outlook assumes depreciation of \$9.7 million, stock-based compensation expense of \$8.2 million and interest expense of \$6.8 million. We also expect to incur between 250 and \$275 million of capital expenditures during the year principally for the Plant 2 project.

Before turning the call over to Don, I want to express that it's been a privilege to work with my fellow Aspen employees since 2006. The team at Aspen is talented, hardworking and completely committed to the company's success. I particularly want to thank Don for his leadership over the years and our shared goal of building Aspen into a great company, rich with opportunity.

In addition, I want to stress how truly impressed I'm with Ricardo. He is remarkably talented and has the right combination of automotive and financial experience to ensure Aspen will succeed in the coming years. With Don at the helm and Ricardo as CFO, I'm confident we will not skip a beat after my retirement.

And finally, I want to thank our investors and analysts. It's been an honor to work with you all since our IPO in 2014.

I'll turn it over now to Don.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [4]

Thanks, John. Good evening, everyone. Thank you for joining us for our Q4 2021 earnings call.

When John and I first met in 2006, I convinced him to join the company to help create a strong foundation. I said just give us three years. Well, of course, over the last 15 years, John has played an instrumental role in building a company that has a vast opportunity and a culture of openness, transparency and fairness.

To honor, John, we as an Aspen team commit to keep this culture at our core. On behalf of all of us at Aspen, thank you, John. I would like to welcome to this call Ricardo Rodriguez, our Chief Strategy Officer, who will replace John as our CFO upon John's retirement at the end of the first quarter.

Ricardo has a deep automotive and Tier 1 background and has already become a valuable member of the Aspen team. Ricardo is another example of the many talented people we have at Aspen, who are turbocharging our efforts to become a highly valuable technology leader in sustainability.

Today, I will describe the key elements of our progress towards achieving our near-term and longer-term business goals. Ricardo will provide a detailed assessment of our thermal barrier opportunity. John, Ricardo and I will conclude today's call with a Q&A session.

The first key point to cover is the recent \$150 million investment from Koch Strategic Platforms. The investment is comprised of convertible notes and common equity and follows KSP's \$75 million equity investment made in June 2021. We appreciate the vote of confidence from KSP and as importantly, the extension of the working relationships we have with various entities at Koch Industries. As I have described in the past, we are leveraging several resources at Koch with the goal of de-risking the scaling of our business as we ramp to be a fast-growing supplier of critical parts to EV OEMs.

The most immediate example relates to the planning, engineering and execution of our Plant 2 capital project. The \$150 million investment from KSP and our year-end cash balance of more than \$75 million

provide a strong foundation for raising additional capital over the next year as we fund our significant growth and value creation initiatives.

The second key point to cover is an update to the growth targets that we first shared approximately one year ago. At that time, we set targets to double revenue from 2021 to 2023 and to double revenue again from 2023 to 2025. We are reiterating the target to double revenue from 2021 to 2023, and we are now upgrading the target to triple revenue from 2023 to 2025 to approximately \$720 million in 2025. We have made significant progress during the past year in order to reach and expand these targets.

In 2021, we generated PyroThin revenue from 10 customers. PyroThin revenue for 2021 and estimated for 2022 is multiples larger than what we expected and discussed one year ago. The doubling of revenue from 2021 to 2023 is based on the expectation that our industrial business will return to pre-pandemic activity levels, which we are well on the way to achieve and that we ramp PyroThin thermal barrier revenue according to existing awarded programs with our North American and Asia-based automotive OEMs.

We estimate that the awarded business from these two customers to whom we are now delivering production parts represents 100% of the 2023 PyroThin thermal barrier revenue target. Like other Tier 1 suppliers, we define the term awarded business as estimated gross program revenues from the volume forecasts of customers taking into account our negotiated program pricing.

The upgraded target to triple revenue from 2023 to 2025 assumes low double-digit growth in our industrial business and anticipated continued penetration in the EV space. We estimate that 60% of the targeted PyroThin revenue for 2025 will be derived from our existing awarded programs. Furthermore, we anticipate that we will convert our highly active prototyping and quoting work into additional program awards in the battery platforms of other automotive OEMs.

The third key point focuses on our plans and expectations for Plant 2 in support of our expanding and accelerating demand plan. We announced today that Plant 2 will be in Statesboro Georgia. Statesboro is a university town near the port city of Savannah and central to a growing automotive and battery ecosystem.

We plan to build Plant 2 considerably larger than initially anticipated, with expected revenue capacity of approximately \$1.35 billion. The first phase of Plant II will have approximately \$650 million of revenue capacity and the start-up date in late 2023. At that time, we will have the revenue capacity in our two plants of approximately \$900 million, enough to service our industrial business and approximately 2.2 million electric vehicles.

As we approach full capacity utilization at this level, we project our gross margin to be approximately 35% and our EBITDA margin to be approximately 25%. During the building of the first phase of Plant 2, we also plan to construct the necessary infrastructure to support both the first and second phases. This approach is most cost-efficient and enables us to add another \$700 million of revenue capacity in approximately 15 months from the kick-off of the second phase. The volume and timing flexibility is critical, as we anticipate potential additional awards for battery platforms from other automotive OEMs.

As a reminder, we estimate we have existing capacity adequate to meet the target to double revenue from 2021 to 2023 to approximately \$240 million. The targeted tripling of revenue from 2023 to 2025 to approximately \$720 million can be met from Plant 1 and the first phase of Plant 2, and points directly to the reason why we are building Plant 2 to be much larger than the original plan.

Upon the completion of the second phase, we expect to have overall revenue capacity of approximately \$1.6 billion across both plants. With the auto industry on track to invest \$0.5 trillion in the next five years to make the transition to electric vehicles, the speed and size of our potential ramp is increasing rapidly. With our technology and scale, we are positioning ourselves to be a valued supplier to a growing list of EV producers. We project the first phase of Plant 2 with full infrastructure will require capital of approximately \$575 million, and the second phase will require capital of approximately \$125 million.

CapEx per dollar of revenue for the full project is less than \$0.55, which is a lower cost than our prior projection of \$0.65 per dollar of revenue capacity. Our year-end cash balance of more than \$75 million and the KSP investment of \$150 million provide the financial foundation for a business able to generate \$1.6 billion of revenue, approximately \$550 million of gross profit and approximately \$400 million of EBITDA.

We will explore over the next year financing alternatives to raise additional cash required to fund the remainder of our significant business opportunity.

In addition to our EV and industrial businesses, we continue to invest in the strategy to leverage our aerogel technology platform into other high-value markets with sustainability themes. Our work with carbon aerogel is the catalyst for Aspen battery materials where our initial focus is on the silicon-rich anode materials.

During 2022, we plan to share key metrics related to performance, cost and third-party validation and to compare those metrics with other interesting companies in the field. With the 2021 investments in people and equipment, ABM is well positioned to respond to the increasing number of requests from battery and automotive OEMs for evaluation materials and for development collaborations.

Finally, I would like to begin the practice of highlighting our ESG work during quarterly earnings calls. ESG has become a business imperative for all good companies. Accordingly, we have begun the process of reorganizing our environmental and social efforts into a more formal ESG strategy. This spring, we plan to publish our first ESG report, which will be followed by a more in-depth report where we will recap our 2022 ESG progress and define our ESG positioning for the future. Our goal is to provide a foundation for deeper discussions on specific ESG topics with investors and other stakeholders in order for various ESG scorecards and rating agencies to have access to accurate and timely information.

And with that, let me now turn the call over to Ricardo. Ricardo?

Ricardo C. Rodriguez [5]

Thank you, Don.

I really appreciate it and couldn't be happier being part of the team. As we execute the plan that Don laid out and invest in increasing our PyroThin capacity to Plant 2, we think that it's worth stepping back and spending some time communicating in detail what makes this such a compelling mission. Thermal runaways and uncontrolled energy released in the cell triggered by various design, manufacturing and usage issues that ultimately result in the decomposition of all of the materials in the cell. A runaway cell's high temperature can push the nearby cell into thermal runaway and such propagation usually destroys the battery or starts a fire.

A lithium-ion battery pack when on fire consumes everything that it can due to its high temperature, including an entire car. These fires usually worsen with water as it reacts with lithium to expand the fire and only Class B type of extinguishers can be used to control them and put them out. The amount of energy released in thermal runaway is significant and equal to more than the energy used to charge the pack, given the formability of some of the materials within the pack.

So, this is one of the most complex system-level issues that our customers face as they integrate and launch safe and reliable electric vehicles in record time. It's really exciting to be enabling a safe and reliable transition experience through electrification as part of some great vehicle nameplates and body styles that we can all safely enjoy with our families. If we put ourselves in our customers' shoes, integrating an EV is no easy exercise. This complexity increases when an OEM takes over the design and integration of the battery pack as well. In this feat of system integration, a vehicle has many diverging requirements or considerations that characterize it, where making one better tends to affect the other. These things determine whether your car is big, small, fast, efficient, agile, full of feel or boring to drive. Things like the vehicle size, weight, range, its performance targets, charging, discharging rates, ease of assembly, crashworthiness, recyclability, et cetera, drive battery pack design and architecture decisions.

These decisions include what cell chemistry, what form factor, pack design, or layout are used, all with the intent of providing as much energy density per unit of mass and space in the vehicle. A key issue is that this energy density needs to stand the test of time and persist within an acceptable and consistent range that consumers see every day on their gauge as they fast charge or drive their vehicles in all sorts of temperatures and conditions for over 10 years.

The integration is very complex even if safety is de-prioritized. However, we are seeing most OEMs go the extra mile prioritizing safety and addressing thermal runaway with both active measures in passive systems

such as our PyroThin thermal barriers. Our value proposition is highest in NMC and mixed silicon graphite anode chemistry. The OEMs that most eagerly reach out to solve their challenges are focused on pouch or prismatic form factors, given their high-energy density and the space for passive protection features as these enable.

The recent news of manufacturing defects, strong impacts, or erratic charge in new cycles causing EV fires are evidence that active systems alone can't prevent thermal runaway 100% of the time. OEM investment in advanced active systems is expensive and time-consuming from an R&D standpoint as their validation requires testing multiple variants of every control strategy, times every type of cell, module and pack to the point of failure. Validating advanced controls for electrochemical processes is hard.

We are encouraged by recent investments from OEMs and other Tier 1s in this space, given that we are all in the first inning of ensuring electric vehicle safety. But our customers foresee a passive system as a key part of their battery pack design in the same way that a vehicle with traction control, automatic emergency braking and collision avoidance systems still requires crumple zones, high-strength steel beams and doors, impact-resistant fuel systems, airbags and seatbelts.

Again, this is a multi-variable problem with varying requirements, customer priorities and approaches. But as we started production and engage in conversations with more customers, the need for passive design elements and materials that work to provide thermal and fire safety as a system is becoming increasingly clear. There is no silver bullet to prevent or stop the effects of thermal runaway, and we intend to work with the rest of the industry to provide a very good lead bullet to this problem for a long time.

We all know that not all materials are alike. And when these materials are fighting for precious space and weight inside of a battery pack to perform a critical safety function, they better deliver and prove their worth. At the core, our customers are looking for the thinnest and lightest material that provides the highest level of runaway prevention and protection. They also want a material that can compress and recover without degradation to keep the cells in their place as their form changes during their life.

On the left, you can see in a simplified way how PyroThin and other materials work together as gas barriers, cell-to-cell barriers and compression pads inside of the pack. They tend to behave very differently on this thermal runaway -- thermal resiliency or fireproofing range, along with the range of force under which they can be compressed and then be able to fully recover.

You can also see that per unit of thickness, it would take twice the material and 2.8x the mass of the closest cell-to-cell barrier material to deliver the performance of PyroThin. These charts also show why increasing the thermal and fire resistance of the material per unit of thickness and weight, along with broadening its compressibility and recovery range are at the heart of our PyroThin R&D efforts.

A thermal barrier that also acts as a compression pad is an example of how we'll optimize content per vehicle or CPV. Beyond that, we will continue to focus R&D on the scalability of our manufacturing processes for the Aerogel base, along with automation tooling and systems for thermal barrier fabrication. On the far right, you can see how we're currently thinking about CPV. Given that we're in the first innings of this type of vehicle content, having just started supplying production parts in the last quarter, we're showing you a broad range of where our supplied CPV lies, whether we're a Tier 1 or Tier 2 supplier.

Our understanding is informed by the vehicle programs that we're supplying and it's evolving with our quoting. What we know for sure is that we won't win by solely focusing on increasing CPV, as we need to earn every cubic inch or ounce inside customers' battery packs. We are focused instead on increasing the percentage of the overall thermal barrier content that is PyroThin. For our projection, we're assuming a CPV that settles in the range of \$300 to \$325, with PyroThin content making up around 60% of that, understanding that this will vary with design changes, new program awards and customers' increasing desire to prioritize safety.

We really look forward to enabling customers accelerating EV product plans and evolving needs with a system-level approach to thermal runaway solutions. Serving as the partner with the right R&D, validation, design, integration and production capabilities gives us the right to play in multiple levels of the value chain, and we will continue delivering accretive CPV. Again, these are our initial thoughts and we will provide updates as our thinking evolves.

With that, I'm happy to turn the call back to Don to summarize our strategy and deliver his closing remarks.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [6]

Thank you, Ricardo.

Before we move to Q&A, let me summarize key points. Our entire focus is on creating significant value with our aerogel technology platform. We are playing in large markets as a technology leader and sustainability. Our talented team is executing our strategy to optimize and expand the value of our current set of markets and investing in leveraging the technology into a next set of opportunities. We received a \$150 million investment from KSP with our -- which with our year-end cash balance of over \$75 million, provides a strong foundation and momentum for raising additional capital over the next year to fund our initiatives to create significant value.

We are reiterating our 2021 to 2023 revenue growth target of 2x and upgrading our 2023 to 2025 revenue growth target from 2x times to 3x times to \$720 million in 2025. We are committed to building Plant 2 to be larger and more capital efficient than initially planned to better match the growing and accelerating customer demand plan and to support the business able to generate an estimated \$1.6 billion of revenue, approximately \$550 million of gross profit and approximately \$400 million of EBITDA.

I will now turn the call back to Lauren for the Q&A session.

Question And Answer

Operator [1]

[Operator Instructions] Our first question comes from the line of Eric Stine from Craig Hallum. Eric, please go ahead.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst [2]

Yes. So great that you could disclose GM and Toyota in the presentation. Just curious, if you could give more clarity on the 10 OEMs that you shipped to in 2021, maybe how that breaks down between the two customers you are shipping to commercial volumes and the other? And then what kind of visibility that potentially gives you into whether it's RFQ activity going forward next awards, things along those lines?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [3]

Well, thanks Eric. We are -- of those 10, let me just say that the fact that we are generating revenue means that we are providing prototype materials to these potential longer-term customers. And we are, I would just say going -- becoming much more expert in this field and a much greater resource to these next set of customers as they develop their battery platforms. And I would just say that those battery platforms through the gating item and being awarded additional programs really is the development of those battery platforms. I would say that the earlier customers that we had were ahead in many regards from the development of those battery platforms. And so we are on a nice pace to add to that list and we believe that we will do that over both 2022 and onwards.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst [4]

Got it. So, I mean, is it fair to say that it's more about the development, I mean, in terms of when we think about timing for more work, it's more about the development on the OEM side rather than you winning the war to making progress within those OEMs?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [5]

Yes. Eric, we think that we have a strong solution that we're bringing real value to these companies. I think Ricardo did a nice job describing it. I would also just reiterate one of my comments, which was that for our 2023 target revenue from PyroThin, we feel that we have 100% of that estimated revenue from those

currently awarded programs. And so we're in a strong position in this first double. And then with respect to the -- to the tripling, 2023 to 2025, we estimate that we have approximately 60% of the revenue from our current awards, again, estimated to play out in that timeframe. So, we're in a strong position. And given that sequence, if you will, we are confident that we will be able to bring on additional battery platforms, additional awards to supplement, let's just say our revenue mix as we get out into that 2023 and 2025 timeframe. But we know, Eric, these builds are right.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst [6]

Yes. No, absolutely. On the topic of 2025, I know you've increased that quite a bit and obviously more OEMs, but maybe if you could just kind of square that or describe that how you have in the past, you've kind of talked about the three levels of engagement. So maybe where that stands now, maybe where it did versus where it did a quarter ago or a year ago?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [7]

We're providing detailed prototype parts to a wide range of these automotive OEMs today. And let me just say a similar track to the one that we are on in the two wins that we've garnered to date and the two awards that we've received. So we're on a very similar track to be able to win additional awards, Eric. And so we have, again, a lot of confidence that we're going to announce additional awards over the course of 2022 and 2023 and round out that revenue mix such that we're serving a good number of these EV producers.

Eric Andrew Stine, Craig-Hallum Capital Group LLC, Research Division - Senior Research Analyst [8]

And I would assume that means that you with Toyota, I mean, you're assuming or is it fair to say you're assuming that you win the whole platform rather than just the one vehicle?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [9]

We think that we're in a strong position to go beyond a single nameplate, a single model and to win a good amount of Toyota's business across its battery platform.

Operator [10]

Our next question comes from the line of Jeff Osborne from Cowen.

Jeffrey David Osborne, Cowen and Company, LLC, Research Division - MD & Senior Research Analyst [11]

Yeah, I was just curious. On Page, what is 9 of the deck, the non-PyroThin revenue, can you talk about what that is? It's 40% at the weighted average content of \$300 to \$325.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [12]

Ricardo, do you want to?

Jeffrey David Osborne, Cowen and Company, LLC, Research Division - MD & Senior Research Analyst [13]

You got 60% down for PyroThin and 40% as non-PyroThin? So, I was just curious. Is that advanced battery materials or is that revenue attributable to the finishing steps?

Ricardo C. Rodriguez [14]

No. So that's actually revenue where we are putting. So if you look at the configuration of the thermal barrier, the bookends of what our thermal barrier looks like are on the most simplified form. It's basically just aerogel shaped to a particular form and sent to the OEM. But then it's most advanced form is actually a sandwich of multiple layers of different materials, very similar to -- pretty much the ones that we show in the

chart on the left on Page 8. And so when we mention other content when speaking of CPV, we're referring to these other purchase materials that are being integrated as part of the thermal barrier assembly.

Jeffrey David Osborne, Cowen and Company, LLC, Research Division - MD & Senior Research Analyst [15]

Got it. That's helpful. I was wondering if you could also just touch on the savings of \$0.10 per dollar of revenue, where that was attributable to relative to the initial expectations? And then how do we think about cost per square meter for Georgia relative to Rhode Island?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [16]

John, do you want to take that?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [17]

Yes. I want to get a little additional detail on the \$0.10. Can you explain that question?

Jeffrey David Osborne, Cowen and Company, LLC, Research Division - MD & Senior Research Analyst [18]

John, as you said that the goal was \$0.65 of CapEx per dollar of revenue and it's now \$0.55. So, I was curious where that \$0.10 came from versus expectations a year or so ago? And then also if you could compare the cost structure per square meter relative to maybe your latest of the three lines in Rhode Island.

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [19]

Yes. So it's a function of multiple -- it really is the improved design that we're deploying in the Georgia plant, ultimately, in terms of its ability to produce a product. We've always said that our second line is more productive than our first in East Providence. And our third, it's 20% more productive than the second line. And so we've been able to -- we haven't -- we deployed the third production line in East Providence in 2015.

And so our designs, our development, our technology, our manufacturing process technology has advanced significantly in that time period. So as we design this plant and we've worked our way through bottlenecks, it is just a far more productive design, more energy efficient, cleaner, greener and more productive. Then in addition, in terms of its use of utilities and a lot of the sort of operational costs, we're seeing operating costs savings of about 33% over the -- what we actually have in East Providence, Rhode Island. So just a more efficient, more productive asset, and it's -- when we designed it and we looked at the productivity of that design, ultimately, it was better than what we had anticipated a year ago.

Operator [20]

Our next question comes from the line of Alex Potter from Piper Sandler.

Alexander Eugene Potter, Piper Sandler & Co., Research Division - MD & Senior Research Analyst [21]

Okay. Perfect. And congrats, John. Thanks for all the help. So, I guess I have a couple of questions on fabrication. Sure. Did you ever put a price tag on that fabrication facility in Mexico? I don't know how material it is, if it really moves the needle in your CapEx guide.

And then maybe as a follow-on to that. What percentage of the time -- I know that you mentioned there's the 60-40 split in your overall CPV calculation. But what percentage of the time do you expect to be just selling sort of pure product out of the facility in Georgia and what percentage of your customers are going to actually ask you to be doing more value-added through the fabrication steps?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [22]

So, I think -- I mean, I'd probably handle both of those questions. So, first, we didn't provide guidance on Mexican fabrication facility at present. It is little less capital-intensive process than what we have deploying

in an aerogel manufacturing plant. And it was much more able to scale it. So as we see revenue growth, as it evolves, we're able to deploy capital, maybe nine to 12 months in advance of the need for that type of -- that facility. So in the aggregate, this opportunity as it increases, we will deploy a significant amount of capital there, but it is scalable and significantly different from what we've dealt with, with the aerogel manufacturing plant in the past.

Then in terms of the content that we're looking at, when we did the content per vehicle in that \$300 to \$325 range, that's us looking at our existing contracts, our award wins and looking at the range of quotes that we've given to prospective customers as well. And so it is our best estimate of where the content per vehicle is likely to shake out through time. However, I think Ricardo talked about this. We do expect it to evolve and these systems will continue to evolve through time. But at present, I'd say it's a function of the 10 customers that we had in 2021, plus all of the other customers who we provided -- we provided responses to in RFI or RFQ situations. And so it's our best estimate for where we think it will shake out long term.

Alexander Eugene Potter, Piper Sandler & Co., Research Division - MD & Senior Research Analyst [23]

That is super helpful.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [24]

I could just add to that to John's [comment]. Yes. Alex, if I could just add to maybe to John's comment. It is a strategic consideration for us that we will consider here in the coming year or two around Tier 1 versus Tier 2. And it is also true that, today, even in the awarded -- awards that we have to date from the programs, in one case, we're very much a Tier 1 and in another case, we are a Tier 2. And I would just -- the advantage for us in the Tier 1 spot for the moment is that we are much more deeply involved as a technology partner in engineering with the customer a solution. And as Ricardo mentioned in his note, it puts us in a much stronger position to increase the aerogel content. I've talked about in the past around this idea of battery performance and safety, our ability to play greater roles in -- within a battery pack is really critical for us and something that we're very focused on. And we think, again, can be significantly value add from our perspective.

Alexander Eugene Potter, Piper Sandler & Co., Research Division - MD & Senior Research Analyst [25]

So all else equal, if you had the choice between Tier 2 type relationship or Tier 1, you would choose Tier 1 as often as you can, is that fair to say?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [26]

What would you say, Ricardo?

Ricardo C. Rodriguez [27]

Yes. I mean I think it's really -- it's -- right now, it's about maximizing the opportunity and supplying as many OEMs as possible without letting go of the deep technical relationship. So frankly, whether they were a Tier 1 or a Tier 2, as I mentioned in my remarks, right, the industry doesn't like paying people for doing nothing in their value chain. And so we just want to earn our position in the value chain. Looking at the current content and the 60-40 split that we have on our planning expectations, that's in essence, OEM signaling to us that they appreciate the value that we add integrating the thermal barrier as a system, right? But on some parts where the configuration looks like just Aerogel, there will gladly be a Tier 2 that has a deep technical relationship on the material side with a material that is able to do what four or five materials on this chart show, right? So, we honestly don't see it as a mutually exclusive situation as long as we're maximizing the opportunity and really solving OEMs problems first.

Operator [28]

Our next question comes from the line of Colin Rusch from Oppenheimer.

Colin William Rusch, Oppenheimer & Co. Inc., Research Division - MD & Senior Analyst [29]

Guys, as you're getting closer to some of these customers and looking at the real needs within these battery packs, are you working on any meaningful efforts to evolve the formula or the recipe for the PyroThin technology to get enhanced performance where it might be able to capture a bit more volume?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [30]

Well, that's a very good question and the answer is certainly yes. When I referenced the fact that we are learning a lot -- that we are becoming quite expert in understanding the phenomenon itself and given the leverage that we know how to play with our own materials and being asked to do more within these systems, I think it sets us up very well, Colin, to be that industry expert that we want to -- that we want to be.

And you know very well that we've installed well over a \$1 billion of material into certain settings and a lot of that has been around passive fire protection. And this is a very logical extension of that understanding of those learnings. And so I think we've been on an accelerated path here and it is what has given us confidence to reiterate our 2x goal in '21 to '23 and upgrade to a 3x revenue growth goal from '23 to '25. It is a function of our understanding of the situation. It is a function of our engagement with a host of the -- what we think will be some of the leading OEMs in the EV space.

Colin William Rusch, Oppenheimer & Co. Inc., Research Division - MD & Senior Analyst [31]

Great. That's helpful. And then in terms of the financing plan for the facility in Georgia, can you talk about how mature your conversations are around into the equipment finance or leverage on that facility as we begin to build it out?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [32]

John, do you want to -- do you want to take that? You know that, John, and I -- and we've been kind of all of the above kind of approach to this over the years. And I think you can pretty well count on that being true again here over the course of the next year or two as we fund our plan. Obviously, as we have a very defined path to profitability, it continues to open lots of opportunities, additional opportunities for us to raise capital and not to mention generating cash of our own here as we move forward through this time period that we've talked about. So -- right, that's the best form of financing you can have. So, that's really what we're focused on, Colin. But we think that we will have opportunities to smartly and creatively fund our growth plans here over the course of the next one, three and five years.

Operator [33]

Our next question comes from the line of Chip Moore from EF Hutton.

Chip Moore, EF Hutton, Research Division - Managing Director [34]

Congrats on the tremendous momentum, guys. John, thanks, and congratulations. It's been great working with you for the past 10 years or so. Just wanted to circle back to that Tier 1, Tier 2 discussion. I think you've been very smart to be OEM agnostic. Obviously, everyone is really worried about supply chains right now. Is there opportunity to accelerate more wins as OEMs look to lock up capacity? Or is that something you think you've seen already?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [35]

Well, we have a very strong team of people, many of who have joined our company in the last one and two years with deep automotive and Tier 1 backgrounds who come from the very same companies that we're engaged with either as automotive OEMs or as Tier 1s. And there -- we are going through qualification processes. Really on a regular basis, we're engaged with experimenting with different products for different battery platforms really on a daily and weekly basis with a whole host of companies.

So, our engagement and I've said this before, I think winning these is -- the gating item is the development of the platform itself. And we believe that we're well-positioned to play an important role in not only in the two companies with whom we're delivering production parts but to a good number of the companies we expect to be leaders in the EV space going forward in all three major regions.

Chip Moore, EF Hutton, Research Division - Managing Director [36]

Understood. Just maybe one more on sort of more near term as we look to Statesboro to start up later next year. You talked about some of the supply chain constraints that you saw this quarter. Looks like the price per square foot had a nice bump. Maybe just big picture thoughts on margins this year and next as we look towards that start-up.

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [37]

Yeah. So. I think those are -- yeah, and a little bit of a path to profitability question. We see in 2022, we are investing significantly in manufacturing resources where we're seeing an increase in manufacturing expenses. As a result, we're setting up facilities that will handle very significant growth in thermal barrier operations in 2023 and 2024, in line with our growth targets, the 2x and the 3x. And so that's depressing gross margins today and it's reflected in our 2022 outlook.

We did mention it during the scripts, but our gross margin expectations in 2022 would be the high single digits because of those -- the addition of those expenses. We're going to go to Monterrey, Mexico, set up a fabrication operation higher -- put in overhead there, significant overhead that will hit our earnings this year, but will set us up for a significant profit growth in 2023 and beyond. After 2022, we would roughly see for every dollar of revenue that we generate, we get about \$0.40 in gross profit, so an incremental gross margin of about 40% on that incremental revenue and about \$0.33 of incremental adjusted EBITDA per dollar of revenue.

So, about a third of that will drop down to the adjusted EBITDA line. And that's that incremental profitability that ultimately results, so when we get out to \$900 million of revenue to build the second plant -- the first phase of the second plant where we'd see gross margins at that point of about 35%, EBITDA margin about 25%. So the low point in terms of margin and profitability is 2022 with revenue growth, we will see very significant improvements in profitability out in '23, '24 and beyond up to those capacity economics that we gave you in our comments.

Operator [38]

The next question comes from the line of Doug Becker from Benchmark Research.

Douglas Lee Becker, The Benchmark Company, LLC, Research Division - Senior Equity Analyst [39]

You mentioned that at full capacity, you'll be generating significant free cash flow to fund future capacity expansion. I was wondering if you could just talk through some of the big moving parts of going from \$1.6 billion of revenue to \$400 million of EBITDA, what type of free cash flow would you be expecting to throw off at that point?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [40]

Yes. So Doug, I mean it's principally -- so CapEx would be -- when we're starting to build the second plant, most of our financing need, our capital need out over the next couple of years is associated with the Plant 2 construction. As Don alluded to, a lot of that's front-loaded, the \$575 million for first phase, \$125 million for the second phase. And then beyond that, we would just have incremental CapEx associated with expanding our fabrication operations, which would be at a significantly lower level than the kind of investment we require for constructing an aerogel plant. So CapEx, maintenance CapEx would not be a significant drag on our operating cash flow and so a lot of that adjusted EBITDA would be available to us as less free cash flow.

Douglas Lee Becker, The Benchmark Company, LLC, Research Division - Senior Equity Analyst [41]

Would you expect to be a cash taxpayer at that level? Or are there NOL?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [42]

Yes. I mean you can do the math pretty quickly, Doug. But we would -- when we break profitable with those incremental margins I just described, very quickly, we'd be adjusted EBITDA positive and start to generate

net income. And we do have a significant NOL that will block our cash tax payments for a period of time. But we've actually run through those in pretty short order. And so when you're talking about being a \$1.6 billion company with \$400 million worth of EBITDA, very significant net income, we would have gone through our NOLs, and we would be a cash taxpayer at that point.

Operator [43]

Next question comes from the line of Amit Dayal from H.C. Wainwright.

Amit Dayal, H.C. Wainwright & Co, LLC, Research Division - MD of Equity Research & Senior Technology Analyst [44]

Congrats on all the progress.

Operator [45]

The next question comes from the line of Jed Dorsheimer from Canaccord Genuity.

Jed Dorsheimer [46]

Back of the bus here. So just a few questions from me. So, I guess the expectation is -- so when we look on Slide 5 for the annual revenue capacity of the \$900 million and the \$1.6 billion, is that in addition or to the \$250 million from East Providence? Or is that including the \$250 million from Providence?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [47]

That includes the \$250 million from East Providence.

Jed Dorsheimer [48]

Got it. And then on the Providence plant, that's never done \$250 million. So has there been a change in terms of yield out of that plant in terms of the ability to kind of double the revenues there because it seems like there's been some breakthrough?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [49]

Yes. So, I'll take that. So Jed, we've always said that our East -- this is before the thermal barrier business. So the capacity of our East Providence plant has been about \$200 million. That's just straight aerogel on the energy infrastructure market. The increase to \$250 million is a combination of the aerogel content that we're selling in the energy infrastructure market, as well as the aerogel content that we'd be using to support the thermal barrier business, plus the additional materials that Ricardo described that we're incorporating into the thermal barrier sandwiches that we're shipping to our customers. So it's aerogel content, plus purchase materials, plus the revenue associated. We do mark up fabrication expenses as well to our automotive customers. And so it's the combination of all of those. So it isn't a productivity improvement. It's really a change in the nature of our business, but it's the same basic aerogel capacity we had before with additional billable items on top of it.

Operator [50]

Our next question comes from the line of Chris Souther from B. Riley.

Christopher Curran Souther, B. Riley Securities, Inc., Research Division - Research Analyst [51]

I just wanted to confirm on the 60% visibility on the 2025's tripling. That includes GM's platform, plus just the one model with Toyota and Subaru. So, you could presumably get closer to 100%, if you won the full platform with Toyota. Is that kind of a good way to think about the current visibility?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [52]

Yes. I mean, I think one of the reasons why we put that statistic, if you will, that estimate in there was we wanted to demonstrate that thing. The -- with an additional one or two wins in other battery platforms, it is -- the math plays out pretty neatly. Let me just say that. And so our ability to achieve those -- the reason we're confident in those targets, the reason that we're increasing those targets is just a function of our engagement with not only the two customers with whom we're delivering production parts, but to a range of additional ones where we are confident that we're very engaged, again, from a prototyping, quoting, working through quality systems, we have very detailed engagement with these additional companies. So, we feel we're in a strong position to support those 2x and 3x target.

Christopher Curran Souther, B. Riley Securities, Inc., Research Division - Research Analyst [53]

Okay. Great. And then of the eight that you shipped PyroThin for revenue last year excluding GM, Toyota, can you talk about the variety of battery chemistries that have been used by those customers, majority are NMC, but are any using NMC or any other chemistries that you're seeing?

Ricardo C. Rodriguez [54]

Yes I mean -- yes, I'm not sure you are familiar with absolutely all eight of them and that level of detail, but take it away, it's even more at a higher level.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [55]

Yes. I mean they're really all prototype parts. So going into too much detail of the specific chemistry that's being used isn't necessarily aligned with what may end up going to production as well, but they're mostly NMC and either pouch or prismatic cells as we've indicated in the remarks.

Operator [56]

Our final question comes from the line of Tom Curran from Seaport Research Partners.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst [57]

Last but certainly not least, I trust. I want to be sure, John, just congratulate you on financially.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [58]

Batting clean-up.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst [59]

Right Batting clean-up. There you go. John, congrats on financially piloting Aspen through every chapter of its development growth over the past 15 years. I really appreciate your detailed guidance, responsiveness and lucid explanations. So, best of luck to you.

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [60]

Thanks, Tom. Appreciate it.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst [61]

So at your total annual revenue target for 2025 of \$740 million, thermal barriers would be expected to contribute 75% or \$540 million. Of the remaining \$180 million, what would be the expected split between energy infrastructure and the battery materials segment?

John F. Fairbanks, Aspen Aerogels, Inc. - VP, CFO, Principal Accounting Officer & Treasurer [62]

Tom, maybe a good way to calculate it would be just to say that we remember back that our pre-pandemic level of industrial was in the range of \$130 million. And we've assumed low double-digit growth from there. So if you add over the course of those three years or four years, if you're putting a 10% or 12% growth rate on that, we're very comfortable with that kind of [earning into that].

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [63]

Yes. And Tom, the Aspen battery materials business, it's a little unclear in terms of what -- how that will be structured and we're going to have to make a decision about whether we license that technology to others and let them produce some materials, or we invest in additional assets to produce those materials ourselves and sell them into the market. And so at present our estimates, we have the cost and expenses and the research and development investment required to fund that initiatives in our models, but we haven't put in revenue assumptions. And we think that that's a conservative approach, but we ultimately are very excited about that business. We think it will generate revenue, but we don't know what form yet. And so on that basis, we've tried to exclude it from all the projections we provided today.

Thomas Patrick Curran, Seaport Research Partners - Senior Analyst [64]

No, that's -- you both helped to answer what I was trying to get at, but it's reassuring to know that it does not assume any revenue contribution from battery materials. So, you do expect to be able to cover that full \$720 million between thermal barriers and the legacy industrial business. And then -- now that you've been able to confirm that the mystery Asian automakers, indeed Toyota, we can home in with our questions about that pacing of potential of that relationship. Could you update us on where they're at with finalizing their own dedicated battery platform to help us with the timing of when you might win more than just the bZ4X? And then would you expect an expansion to any kind of platform-wide contract to also include Subaru?

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [65]

We just want to be very careful about, talking about specific programs right now and specific expectations with these companies. And we just want to be very mindful and respectful of their own developments and what have you. So let me just say maybe what I've said a bit in the past, which is that we feel we're in a very strong position that these companies, again, those we've been awarded business and others that were in the later stage innings, if you will, of being awarded business we anticipate. I would -- those -- we just feel like we're in a strong position and we feel that we are bringing a lot of value to the discussions.

And it's -- yes, it comes from our technology, but it also comes from the strategy that we've talked about for over a year now around being OEM agnostic, being really the industry resource here, bringing expertise into this really difficult challenging problem. I thought Ricardo did a nice job just describing the complexity of this and the likelihood of these automotive OEMs trying to manage this situation, both with active and passive systems simultaneously. That's the way we see it playing out certainly over any period of time. So, again, we just want to be a little careful about talking about specific programs.

Operator [66]

Okay. That is the end of the Q&A session today. So, I will now hand you back over to Don Young for closing remarks.

Donald R. Young, Aspen Aerogels, Inc. - President, CEO & Director [67]

Thank you, Lauren, very much. To everyone, we appreciate your interest in Aspen Aerogels. We look forward to reporting our Q1 2022 results in late April. Be well. Have a good evening. Thank you very much.

Operator [68]

This concludes today's call. Thank you for joining. You may now disconnect your lines.